Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/03/2024
PI Name:	Robinson, Stephen K. Ph.D.		
Project Title:	A Haptic Sensory Supplement to Optimize In-Flight Adaptive Training for Human Control of Spacecraft Robotic Arms		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
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No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/01/2027 per A. Beitman/JSC (Ed., 8/24/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Joiner, Wilsaan Ph.D. (University of California, Davis) Schofield, Jonathon Ph.D. (University of California, Davis)		
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Task Description:	Addition to Task Description (Ed., 8/25/24) To take advantage of the NASA Human Exploration Research Analog (HERA) C8 schedule, we propose an additional Aim 4: Test to what degree of expertise can subjects learn to use haptic guidance to manually control the robot arm end-effector through a prescribed, but unseen trajectory. In other words, we are performing a sensory substitution from visual to haptic feedback for manual trajectory guidance – something that has never been attempted for space robotics. Aim 4 hypothesis: Through initial training followed by repetitive practice, subjects can become proficient at manual control of a robot arm with trajectory guidance provide by haptic sensory input alone. Note that we do not propose that future space robotics tasks would be performed without visual sensory feedback; rather, we are probing how far haptic sensory signaling can be taken in terms of developed training motor-control skills for multi-dimensional tasks, with robot arm trajectory guidance as an example of a complex manual control task. Background: Human-guided robotic arm operations will be essential for long-duration human spaceflight, including low-Earth orbit (LEO), cisLunar and Martian orbits, as well as for surface support of long-duration habitats. During these missions, astronauts will be tasked with mission-critical and demanding skill-based robotic operations long after their performance peak during pre-flight training. Therefore, they will need to re-acquire proficiency while also adapting to the differences between Earth training and mission operations. Challenges arise with long-term isolation and the accompanying somatosensory alterations during spaceflight. Thus, the astronaut requires onboard training that not only targets skill re-acquisition but also serves as a countermeasure to neuroplastic changes and the new sensorimotor-control environment of microgravity. Finally, human-guided robotic operations involve a potentially overwhelming array of visual feedback elements that must be monitored to acquire and maintain a mental model of the robotic arm, vehicle surroundings, and near-term trajectory projections. Sensory overload is never far from reality and when risking structural collision or injuring an arm-mounted extravehicular activity (EVA) astronaut, and even brief dropouts in visual scan, mental-model updates of trajectory prediction, or situation awareness can rapidly lead to mission-critical safety impacts. Hypotheses: The haptic sensory channel is underutilized in mission-critical robotic operations, especially in microgravity where touch to the body is minimal and even clothing floats freely. Here, we propose to both supplement and relieve the overtaxed visual feedback channel with haptic feedback to improve operational accuracy, task efficiency, and crew safety. Our contribution would serve as an essential element of the new NASA Human Research Program (HRP) Mars Adaptive TRaining-Integrative Knowledge System (MATRIKS) concept. We hypothesize that integrating haptic and visual feedback can: (1) Provide a novel adaptive training strategy in which levels of haptic and visual feedback are linked to in-flight training performance to more rapidly bring a long-duration astronaut to task-ready status; (2) Reduce an astronaut's cognitive burden during mission-critical robotic arm operations, thus reducing opportunities for loss in situation awareness and related operational errors, resulting in enhanced safety; and (3) Promote a stronger sense of tool/motor embodiment to accelerate neuroplastic learning.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are addressing the NRA's proposal Topic Focus #1 by developing a class of in-flight and planetary-surface robotics adaptive training to achieve: - Adaptive acquisition of knowledge during long-duration missions (just-in-time training) - Consolidation of motor skills (transfer of training from on-board simulators to mission tasks) - Retention of motor skills (refresher training and retention assessment) - Meaningful and motivating training regimens (mission-relevant on-board training that is both appealing to the crewmembers and worth the expenditure of valuable crew time) Although robotic on-board training is currently available on ISS via the ROBoT SSRMS simulator, there is no existing spaceflight robotic training technology that is aware of and adaptive to the skill state and learning rate of individual astronauts. Thus, we propose to attenuate HRP risks by addressing somatosensory changes during long-duration missions through augmented sensory feedback to astronauts undergoing onboard just-in-time or refresher training, using experimentally-validated methods that adapt to the assessed training needs of the crew. If our research aims are achieved, we can deliver to NASA a training protocol for deep-space robotics operations that is adaptive to the current skill-state of the astronaut. By delivering a fully-trained and confident robotics operator to the surface of Mars, NASA's mission objectives and safety requirements can be met. In addition, the proposed research as an element of the MATRIKS VNSCOR is expected to lead to improvements and clarifications in NASA STD 3001a, Vol. 2, Rev.B, Part 10.6.1: "Automated and Robotic System Design".
	[NOTE: Per the Principal Investigator, this project is also known internally as, "Morpheus: A Haptic Sensory Supplement to Optimize In-Flight adaptive Training for Human Control of Spacecraft Robotic Arms" (Ed., 1/27/24).] Morpheus Year 3 Progress at University of California (UC) Davis (3 professors): a. Human/Robotics/Vehicle Integration and Performance Laboratory (HRVIP) Lab (Prof. Stephen Robinson): https:// Efforts in the HRVIP Lab since the Year 2 report have focused on developing all aspects of manual control of the UR5e robot arm. These include interfacing with the Trossen arm progress in the Joiner Lab and preparing to incorporate the haptic sleeve being developed mostly in the Schofield Lab. We have begun a migration from ROS1 to ROS2 robotic controlling software and are developing a full graphical simulation of the arm environment to test against the 3D environment. Since we are employing haptic signaling as a proximity warning to potential collision hazards, we needed to develop software that computes and displays in real-time the shortest distance between any element of the work environment and the moving robot arm – this capability is now complete. We are constructing a rapidly reconfigurable robotic work environment based on a 1/20 model of the International Space Station (ISS) – this environment presents a potential "obstacle course" for the arm operator to traverse while avoiding collisions through haptic alerts via our haptic sleeve. Manual control of the UR5e robot arm is via two methods, which we will compare as part of our subject testing: Method 1: Subjects accomplish 6-DOF commanding via dual joysticks, one for 3-DOF end-effector translation, and the other for 3-DOF end-effector rotation/pose. Commanding all 6 degrees of freedom simultaneously is quite difficult, but

Task Progress:	generally is not required to execute the required trajectory. Method 2: We have adapted the “Gello” robotic controller concept developed by UC Berkeley (https://). This controller provides a highly intuitive and easy means to control all 6 degrees of freedom of the robot arm by manipulation of a small, custom-fabricated robot arm that gives virtually real-time motion mirroring of the controller arm by the UR5e arm. We have constructed two of the Gello controllers and are testing them for use in the human subject testing ahead. b. BEAR Lab (Prof. Jon Schofield): https:// We have developed, prototyped, and fabricated multiple revisions of our haptic sleeve system. The sleeve can be worn on the user's upper limb and conveys relevant robotic position and movement information through vibration-based haptic feedback. This sleeve utilizes 34 linear resonant actuators (LRAs), each independently controlled for vibration frequency and amplitude. Presently, we are conducting perceptual testing experiments to determine optimal vibration parameters (frequency, amplitude, waveform, timing of vibrotactors) for effectively encoding directional information to the user, leveraging apparent tactile motion illusions. In this setup, users experience directional waves of vibration traversing their skin through strategic manipulation of vibrotactor activation timing. Simultaneously, we are collaborating with the UC Davis Design Department to create a fabric-based, washable sleeve design that ensures comfortable wear and maintains consistent vibrotactor positioning during donning and doffing. The haptic sleeve is controlled using a combination of an Arduino Portenta and a Raspberry Pi 4. It interfaces with Robot Operating System (ROS) software, facilitating communication of directional movements from our Trossen robotic arm to the user via the haptic sleeve. We are also collaborating with the Joiner and Robinson labs to extend our system beyond the Trossen robotic platform, integrating it with virtual and physical versions of the UR5e robotic arm in their facilities. Additionally, members of the Schofield lab are closely collaborating with the Robinson lab to develop algorithms for robotic arm collision detection. This ongoing work currently utilizes ROS to compute distance vectors between objects and the robotic arm, employing virtual clones of objects within the robotic arm’s physical workspace. c. Joiner Motor Lab (Prof. Wil Joiner): https:// We have successfully investigated the factors that improve the rate of learning to control a robotic arm and the retention of this information by examining the change in key performance and operational measures as a function of training duration. To assess these behavioral metrics, we have formulated a task that consists of moving the Trossen robot's gripper around different barriers in a custom obstacle setup to retrieve a goal object with the following paradigm blocks: 1. During an initial familiarization block, subjects become familiar with controlling the robotic arm using a PS4 controller. 2. During the next baseline block, participants are instructed to retrieve the goal object and transport it to the target area with no obstacle presented. 3. Following the baseline block, participants are trained to perform the same retrieving task with three difficulty levels based on different obstacle placements: (i) Navigate under one bar obstacle 15 cm from the ground level, (ii) Navigate under the same bar obstacle and avoid a second post 5 cm away from the goal object, (iii) Navigate under the same bar obstacle and between two posts 18 cm apart, one which is 5 cm away from the goal object. As a first step, we quantified skill learning by the change in completion time—the movement onset of the robotic arm to the time when the goal object was placed in the target area. In our preliminary data, our participants quickly learned to control the robot to retrieve the object with a 70% decrease in completion time when there was no obstacle (condition i). The participants required significantly more time to complete the task when obstacles were presented (conditions ii and iii), but the completion time was gradually reduced as a function of training duration. Going forward we plan to quantify performance using more operational measures (e.g., trajectory accuracy, smooth motion, error rates, number of collisions/near misses, workload/cognitive margin). In addition, we plan to develop a similar obstacle course for the UR5e Robot and quantify the transfer of the learned skill controlling the Trossen robot to control the UR5e robot by examining the changes in the above performance measures. Finally, we plan to investigate the factors (e.g., training approach and real-time feedback) that improve the rate of learning to control the novel robotic arm and the retention of learning. Overall, we plan to use these measures to determine an optimal training regimen for NASA’s ROBoT/SSRMS training system. Summary of Accomplishments in Year 3: • Completed the UC Davis SmartHab mockup to house Morpheus robotic experiments • Design, fabrication, and extensive testing of a family of haptic sleeves with 34 integrated vibration actuators • Extensive discovery of effective vibration parameters and patterns to be used for coded commanding to the arm operator through haptic signaling on the right arm • Developed reconfigurable “obstacle course” environments for testing the ability of subjects to avoid collisions while manually commanding the robot arm with haptic signaling for impending collision warning • Developed manual control capability and obstacle course environments for both robot arms to be used in the HERA study (Trossen and UR5e) • Development of dynamic proximity assessment software that constantly computes distances between the robot arm and its surroundings as it moves under manual command • Developed and tested two different 6-DOF control methods for commanding robot arm motion • Developed extensive software to enable all of the above capabilities • Completed several human-subject pilot studies to test our technology and procedures • IEEE paper: “A Wearable Haptic Interface for Spacecraft Robotic Arms that Leverages Tactile Stroke Illusions” • International Symposium on Wearable Computers 2024 paper: “A Wearable Haptic Sleeve Interface for Spaceflight Robotic Arm Operations”
Bibliography Type:	Description: (Last Updated: 06/02/2026)
Papers from Meeting Proceedings	Hong K, Zhang R, Lee KG, Young PR, Joiner WM, Robinson SK, Goncu-Berk G, Schofield JS. "A Wearable Haptic Sleeve Interface for Spaceflight Robotic Arm Operations." 28th annual International Symposium on Wearable Computers, Melbourne, Australia, October 5-9, 2024. 28th annual International Symposium on Wearable Computers, Melbourne, Australia, October 5-9, 2024. ISWC paper. , Oct-2024
Papers from Meeting Proceedings	Hong K, Zhou W, Joiner WM, Robinson SK, Schofield JS. "A Wearable Haptic Interface for Spacecraft Robotic Arms that Leverages Tactile Stroke Illusions" IEEE Haptics Symposium, Long Beach, CA, April 7-10, 2024. IEEE Haptics Symposium, Long Beach, CA, April 7-10, 2024. , Apr-2024